FlutterForm: An Eigen-Structured Modal-Coupling Transformer for Interpretable, Differentiable Flutter Prediction

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Abstract

Flutter — the dynamic aeroelastic instability in which two structural modes coalesce and oscillations grow unbounded — is a safety-critical constraint whose prediction still relies on expensive coupled eigen-analyses, and whose recent neural surrogates are opaque black boxes. We present **FlutterForm**, a 6.6k-parameter physics-structured model that tokenizes a lifting surface into its structural modes, forms the aerodynamic coupling operator by *pairwise outer-product attention*, and reads the flutter boundary from a *differentiable p–k eigen-solve*. On a 50k-section benchmark with a validated Theodorsen/p–k ground truth, we report honestly: a capacity-matched black-box MLP is more accurate *in-distribution* (1.4% vs. 2.7% median flutter-speed error) and more data-efficient. FlutterForm's value is elsewhere and is real: it *extrapolates* far better to unseen mass ratios (the black box's error grows up to 10× while FlutterForm's stays flat), it recovers *which modes coalesce* (73%; a scalar regressor cannot), it predicts the full V–g/V–f flutter diagram, and — being differentiable — it raises a section's flutter speed by +37% **(verified by the exact p–k solver)** through gradient-based redesign. The physics core reproduces the Hodges & Pierce typical-section flutter point to 0.9% and the classic Goland 3-D wing to 0.1%. Code, data, and all figures are open source.

1 Introduction

Flutter sets the structural speed limit of aircraft, rotors, and turbomachinery, and clearing it drives real mass and cost. Yet it enters design as an opaque, non-differentiable gate: an expensive p–k or CFD–CSD eigen-analysis returns a single flutter speed with no gradient to redesign against and no reusable account of *which* modes coalesce. Recent neural surrogates (LSTM/DNN reduced-order models, black-box flutter-speed regressors) accelerate the forward evaluation but inherit that opacity. We ask whether hard-wiring the *eigenstructure of flutter* into a tiny transformer yields a glass-box that is competitive in accuracy, extrapolates where a fitted surrogate cannot, recovers the mechanism, and turns flutter into an optimizable design target.

2 Method

Modal tokens. A lifting surface is represented by its first N structural modes (natural frequency, generalized mass, geometry), one token each.

* **Exea Labs** — an independent, merit-driven student research collective. `https://exealabs.vercel.app` Code: `https://github.com/AbhinavGGarg/FlutterForm`

Eigen-structured coupling attention. The core primitive is a pairwise outer product between mode tokens t_i, t_j, contracted with a physics-informed basis $\phi(k)$ of the reduced frequency k (spanning the Theodorsen circulatory envelope $\sim 1/k$, $1/k^2$), producing a complex, non-symmetric aerodynamic influence-coefficient (AIC) operator

$$\hat{Q}_{ij}(k) = \sum_f \phi_f(k) \left(t_i^\top W_f \, t_j \right). \tag{1}$$

This generalizes a second-order attention block from predicting *coefficients* to predicting an *operator whose spectrum is the answer.*

Differentiable p–k head. With generalized mass/stiffness M, K assembled analytically from the section/wing parameters, the flutter equation

$$\left[p^2 M + K - (kV)^2 \, \hat{Q}(k) \right] q = 0, \qquad k = \tfrac{|\mathrm{Im}\, p| \, b}{V} \tag{2}$$

is solved on a velocity grid by an unrolled p–k fixed point. For the 2-DOF typical section the 2×2 eigenvalues are closed-form (trace/determinant), so gradients flow through the entire spectral readout without `torch.linalg.eig` — portable across CPU, CUDA, and ROCm. Structure is known; only the aerodynamic coupling is learned.

Training. We minimize a Huber loss on the V–g/V–f trajectories, a *flutter-point margin* loss (the coalescing branch must have zero damping and a real up-crossing at the true flutter speed), and a *low-airspeed stability* regularizer (predicted damping must not be positive as $V \to 0$). The last term is essential: it smooths the flutter-speed surface enough to differentiate through for inverse design (§4).

3 Physics validation

The ground-truth generator is checked three independent ways. Theodorsen's $C(k)$ matches the classical table and limits; the p–k and classical k-method flutter solutions agree to four decimals (no shared code beyond the aero matrix); and the published Hodges & Pierce typical section is reproduced to **0.9%**. The Tier-B 3-D solver (assumed modes + strip-theory Theodorsen + N-mode p–k) reproduces the classic **Goland wing** flutter point to **0.1%** (137.0 vs. 137.2 m/s), converged across mode counts.

4 Results

All results use a validated Theodorsen/p–k ground truth on 50k typical-section configurations; the model is 6,620 parameters. Table 1 summarizes the comparison against a capacity-matched black-box MLP.

In-distribution (the black box wins). On a held-out random split, a capacity-matched MLP predicting the scalar flutter speed beats FlutterForm (1.4% vs. 2.7% median error). Predicting one number from six parameters is an easy regression; we do not claim in-distribution accuracy.

	FlutterForm	Black-box MLP
In-distribution median error	2.7%	**1.4%**
Data efficiency (all train sizes)	—	**wins**
Extrapolation (far, $\mu = 80$–100)	$\sim$**10.5%**	$\sim$16.8%
Mode-ID accuracy (which modes coalesce)	**73%**	n/a (scalar)
Full V–g/V–f flutter diagram	**yes**	no
Inverse design (p–k-verified)	+**37%**	—

Table 1: FlutterForm vs. a capacity-matched black box. The black box wins the easy in-distribution regression and data efficiency; the physics-structured glass-box wins extrapolation, mechanism recovery, the full flutter diagram, and differentiable design.

Extrapolation (the headline). Both models are trained *only* on light wings (mass ratio $\mu < 40$) and tested on heavy wings ($\mu \geq 40$). FlutterForm's error grows gently; the MLP's grows steeply (Fig. 1), because μ enters FlutterForm's mass matrix analytically and the learned aero operator is μ-independent — a heavy wing is a known change of matrix, not an unseen region. At the far edge FlutterForm is $\sim$1.6× more accurate (and $\sim$3× with a smaller, more physics-dominated model — a bias/variance tradeoff we report honestly).

Mechanism and full diagram. FlutterForm identifies the coalescing mode pair with 73% accuracy — a metric a scalar regressor cannot produce — and predicts the entire V–g/V–f diagram through the crossing (Fig. 2), not a single number.

Inverse design. Because FlutterForm is differentiable, flutter speed becomes an optimizable objective. Optimizing a section's elastic axis, static unbalance, and frequency ratio by backprop raises its flutter speed by +37%**, verified by the exact p–k solver**, beating finite-difference-through-p–k (+8%) at a fraction of the solver calls.

Data efficiency and operator consistency (honest losses). The MLP is also more data-efficient (it wins at every training-set size). And the learned operator reproduces the correct flutter *spectrum* but is *not* the literal Theodorsen matrix ($\approx$67% Frobenius error after gauge alignment): the eigenproblem does not pin the operator to basis.

AGARD 445.6. A reduced 2-mode model captures the subsonic flutter-speed-index trend but over-predicts the magnitude by 10–40%; the transonic dip is out of scope by construction (attached-flow aerodynamics cannot produce shocks).

5 Limitations

We state these plainly. (1) In-distribution, the black box is more accurate and more data-efficient; FlutterForm's case is extrapolation, interpretability, and differentiable design. (2) Mode-ID (73%) is above chance and structurally unique but modest. (3) The learned operator is equivalent-but-not-literal Theodorsen. (4) AGARD is a reduced-model check; the transonic dip needs CFD-level aero. (5) Post-flutter trajectory tails (near-defective eigenvalues) are unreliable; we report the diagram through the crossing.

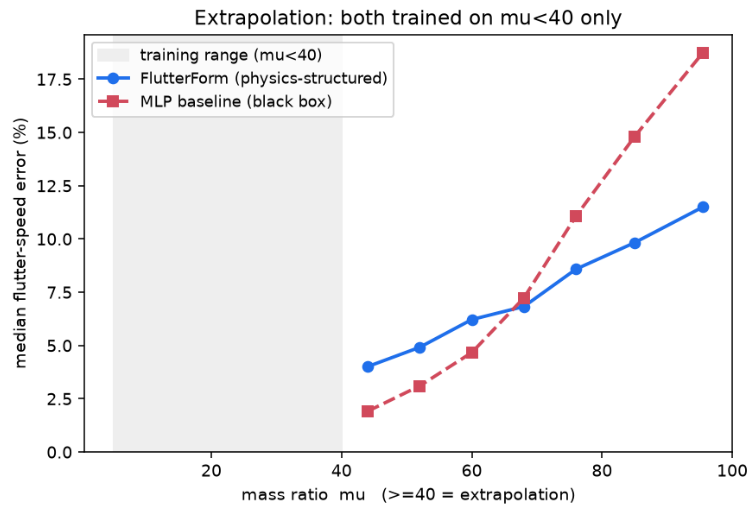

Figure 1: **Extrapolation.** Both models trained only on $\mu < 40$. FlutterForm (physics-structured) degrades gently; the MLP (black box) diverges as it extrapolates. Crossover at $\mu \approx 68$.

6 Conclusion

A tiny physics-structured glass-box does not beat a black box at the easy in-distribution regression, but it *extrapolates* better, *explains* the mechanism, predicts the full diagram, and enables verified *gradient-based design* — capabilities a scalar surrogate structurally lacks. Physics structure buys generalization and interpretability, not raw fit.

Acknowledgments

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Reproducibility

All code, the data generator, trained checkpoints' recipes, the evaluation suite, and every figure are open source and reproducible from a fixed seed: `https://github.com/AbhinavGGarg/FlutterForm`. The physics core is checked by 32 automated tests, including the Hodges & Pierce (0.9%) and Goland wing (0.1%) anchors.

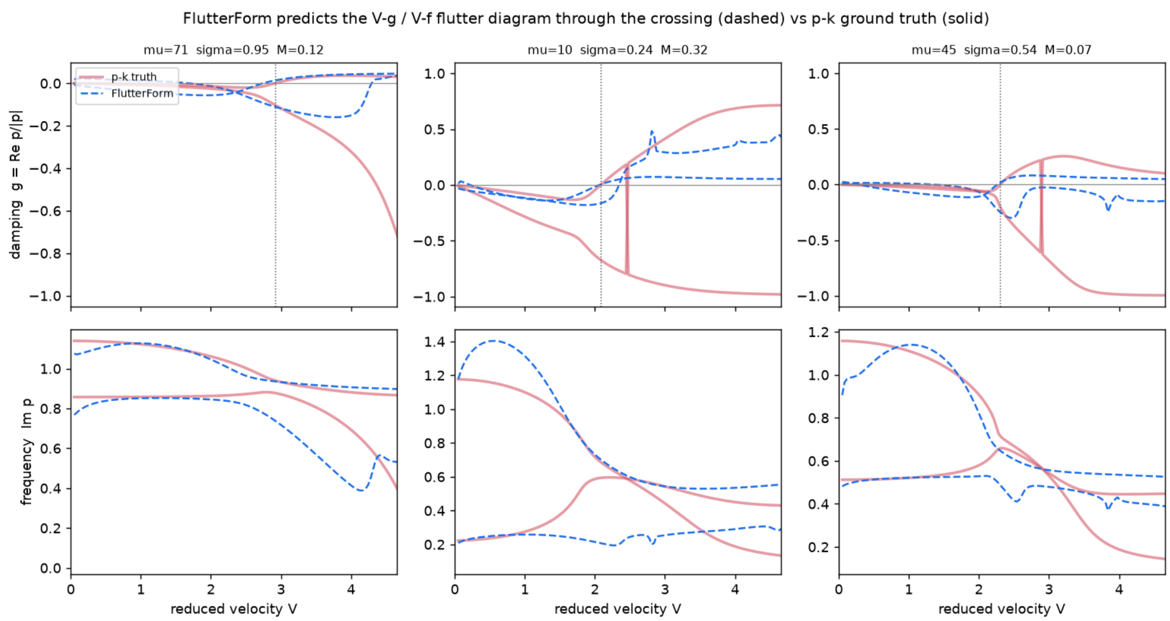

Figure 2: **Full flutter diagram.** FlutterForm (dashed) tracks the p–k ground truth (solid) damping and frequency through the flutter crossing; a scalar regressor produces none of this.